

USER MANUAL

Inventory Management Application (IMA)

Course Code: CSD026

Department: Computer Science

Platform: Java + JavaFX + MySQL

1. Introduction

The **Inventory Management Application (IMA)** is a desktop-based system designed to manage and track departmental inventory efficiently. It replaces manual, paper-based record keeping with a centralized digital solution. The system is intended for **Inventory Managers, HOD, and Dean**, each having role-based access to system features.

The application helps users manage products, suppliers, purchase bills, issuance, returns, and reports through an easy-to-use graphical interface.

2. System Requirements

Hardware Requirements

- Minimum 4 GB RAM
- Standard keyboard and mouse
- Display resolution: 1024 × 768 or higher

Software Requirements

- Operating System: Windows 10 or higher
 - Java SE 17 (LTS)
 - JavaFX libraries
 - MySQL Database Server
 - NetBeans IDE (recommended)
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3. User Roles

Inventory Manager

- Add, delete, and modify inventory records
- Issue and return items
- Generate reports
- Track usage
- Request new inventory

HOD / Dean (Administrator)

- View reports
 - Approve inventory requests
 - Monitor budget analysis
 - Track faculty-wise usage
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4. Login Procedure

1. Launch the Inventory Management Application.
 2. Enter **Username** and **Password**.
 3. Click the **Login** button.
 4. If credentials are valid, the dashboard screen opens.
 5. If credentials are invalid, an error message is displayed.
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5. Dashboard Overview

After successful login, the **Dashboard** appears.
It contains the following buttons:

- **Update**
- **Issue**
- **Return**
- **View**
- **Report**
- **Budget Analysis**
- **Track Usage**
- **Request Inventory**

At the top-right corner:

- **User Manual:** Opens the help document
 - **Logout:** Ends the session securely
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6. Update Module

Clicking **Update** opens a dropdown with the following options:

6.1 Add

Used to add new inventory details.

Fields include:

- Supplier Name
- Email
- Address
- Contact Number
- Product Type ID
- Product Name
- Product Description
- Bill Invoice
- Bill Received By
- Date of Purchase
- Quantity Ordered

Click **Upload** to attach bill documents.

Click **Back** to return to the dashboard.

6.2 Delete

Used to remove inventory records.

Fields include:

- Product Type ID (dropdown)
- Product Name (dropdown)
- Product ID (dropdown)

Users can select values from dropdown lists instead of typing.
This reduces errors and improves usability.

6.3 Modify

Allows updating existing records.

Modify options include:

- Product
- Supplier
- Bill

The user selects the category and updates required details.

7. Issue Module

Used to issue items to staff or departments.

Fields include:

- Product Name
- Issued To
- Issued By
- Department Name
- Quantity Issued
- Reason
- Issued Date

The system checks available stock before issuing items.
If stock is low, a **warning alert** is shown.

8. Return Module

Used to record returned items.

Fields include:

- Product Name
- Return To
- Return By
- Quantity
- Date

Returned quantity is automatically added back to available stock.

9. View Module

Allows users to:

- View product records
 - Search inventory items
 - Filter records based on criteria
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10. Report Module

Clicking **Report** opens a dropdown with:

- **Weekly Report**
- **Monthly Report**
- **Yearly Report**

Reports summarize:

- Stock availability
 - Issued items
 - Returned items
 - Usage trends
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11. Budget Analysis

Provides a financial overview of:

- Inventory purchases
- Department-wise expenses
- Comparison between periods

Accessible to HOD and Dean.

12. Track Usage

Displays faculty-wise or department-wise inventory usage.

Helps administrators monitor consumption patterns.

13. Request Inventory

Inventory Managers can raise requests for:

- New items
- Stock replenishment

Requests are forwarded to:

- **HOD** for approval
 - **Dean** (for large or computer-related purchases)
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14. Alerts and Notifications

- Low inventory alerts

- Validation error messages
 - Confirmation dialogs for delete actions
 - Success messages after operations
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15. Logout

Click **Logout** from the dashboard to:

- End the current session
 - Return to the login screen securely
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16. Conclusion

The Inventory Management Application provides a structured, reliable, and user-friendly solution for managing departmental inventory. By automating inventory operations, it improves accuracy, transparency, and efficiency while reducing manual workload.